Fiscal Year:	FY 2019	**Task Last Updated:**	FY 06/10/2021
PI Name:	Stamper-Kurn, Dan Ph.D.		
Project Title:	Coherent Magnon Optics		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	dmsk@berkeley.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	510-642-9618
Organization Name:	University of California, Berkeley		
PI Address 1:	Physics		
PI Address 2:	366 Leconte Hall 7300		
PI Web Page:			
City:	Berkeley	**State:**	CA
Zip Code:	94720-7301	**Congressional District:**	13
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2013 Fundamental Physics NNH13ZTT002N (Cold Atom Laboratory--CAL)
Start Date:	04/01/2014	**End Date:**	04/30/2019
No. of Post Docs:	3	**No. of PhD Degrees:**	2
No. of PhD Candidates:	7	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JPL
Contact Monitor:	Israelsson, Ulf	**Contact Phone:**	
Contact Email:	ulf.e.israelsson@jpl.nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Mueller, Holger Ph.D. (University of California, Berkeley)		
Grant/Contract No.:	JPL 1505588		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The spin degree of freedom of ultracold atoms makes the physics of quantum gases richer, and offers opportunities for fundamental science explorations using cold atomic media. The scientific agenda we propose for the Cold Atom Laboratory (CAL) builds upon on our world-leading research on spinor Bose-Einstein condensates, including our development of magnetization-sensitive imaging and studies of spin coherence in optically trapped quantum gases. We pursued four major activities: (1) Precise characterization of magnons in spinor Bose-Einstein condensates: Magnons are low energy spin excitations in spinor condensates. Theories of symmetry breaking, as well as mean-field and beyond-mean-field mean-field treatments of Bose gases provide strong predictions for the dispersion relation, mass, and excitation gap of magnons. These were tested experimentally. (2) Magnon interferometry in dense Bose condensed gases: magnons in ferromagnetic spinor condensates, such as the F=1 gases of 87Rb and 41K, are by necessity gapless and quadratically dispersing, with an effective mass that, within mean field theory that is applicable to weakly interacting gases, is equal to the atomic mass. In other words, magnons propagate just like free particles. We pursued the possibility of implementing atom interferometry based on coherent magnon optics. Compared to a Bragg interferometer, a magnon interferometer has no mean-field interaction shift. Magnon interferometry was realized and investigated. (3) Magnon-evaporative cooling: We explored a novel cooling technique in which magnon excitations are used to absorb, and then expel entropy from a highly degenerate spinor Bose-Einstein condensate. Our target was to produce and also confirm the production of quantum gases at the lowest entropy per particle, with a method that can be applied in the CAL apparatus. (4) Quantum gas mixtures: A unique capability for quantum gas experiments in microgravity is that of overlapping quantum gases composed of different atomic species within the exact same trapping potential. On Earth, trapping requires one to overcome the force of gravity by applying electromagnetic fields to the trapped atoms. The response of each element to electromagnetic fields is different, meaning that it is impractical to place a quantum gas composed of several different atomic elements within the exact same potential. Aboard CAL or other follow-on missions, quantum gases can be admixed within potential free volumes without needing to apply an compensation against gravity. As preparation for such free-fall experiments, we explored the nature of quantum gas mixtures of lithium and rubidium atoms. Our goal here was to characterize the interaction between these gases, including the spin dependence of such interactions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research is fundamental in flavor. We are exploring new phenomena and new experimental capabilities in a physical system that is itself new and only recently available for experimentation. Exploring in this new space of questions has prospects for making discoveries. More specifically, the research impacts our knowledge of materials (through studies of superfluid turbulence), extends the frontiers of natural phenomena that we can access and explore (through the development of new thermometry and cooling techniques), and contributes to the development of high precision sensors for future scientific and technological applications (through studies of atom and magnon interferometry).
	Referring to the four major activities listed in the task description: (1) Precise characterization of magnons in spinor Bose-Einstein condensates: We focused on the nature of magnons within F=1 spinor gases of 87-Rb atoms. These gases are ferromagnetic in nature, meaning that interactions between atoms favor a state in which all atoms are magnetized, with the magnetization oriented in spatially uniform direction. Magnons are the magnetic excitations of such ferromagnetic states, i.e., a spatial ripple in the uniform magnetization. These excitations have a characteristic excitation energy that varies with the length-scale of the magnetization ripple (the wavelength of the magnon). While in many solid magnetic systems the energy of magnon excitations (i.e., their dispersion relation) is difficult to calculate from first principles, here, in the case of atomic quantum gases, there are robust theoretical predictions for the magnon energy. We conducted experiments to test these predictions. The tests were reported in two prominent publications: a study of coherent magnon propagation [Marti et al, Physical Review Letters 113, 155302 (2014)] and a study of the phenomenon of magnon condensation [Fang et al., Physical Review Letters 116, 095301 (2016)] [Ed. note: see Cumulative Bibliography] In the former, we quantified the magnon dispersion relation in two separate measurements. In the first, we studied the energy of uniform magnon excitations. For this, we examined a spinor Bose-Einstein condensate with a slowly varying atomic density. After exciting long-wavelength magnons in the system, via a uniform rotation of the condensate magnetization, we detected a spatial variation in the evolution of the excitation. From this, we determined that the uniform magnon excitation has an energy cost that varies with the gas density. We ascribed this energy offset (known as the magnon energy gap) to the magnetic dipole-dipole interactions in the atomic gas. By this measurement, we obtained the first direct quantitative evidence of magnetic dipolar effects in alkali-atom quantum gases. Another measurement reported in the 2014 paper focused on the magnon recoil energy, which is the added energy required to boost a magnon excitation from zero momentum to finite momentum. This recoil energy was measured through magnon interferometry, which we describe further below. The dispersion relation was found to be particle-like, characterized by a magnon mass that was slightly but significantly different from the atomic mass. There remains some theoretical controversy over this measurement result. In our second paper, we studied the thermodynamic behavior of many magnon excitations evolving within a ferromagnetic Bose-Einstein condensate. Magnons are bosons. When a thermal gas of magnon excitations becomes sufficiently cold and dense, these magnons should themselves undergo Bose-Einstein condensation. Such condensation results in a phase transition of the ferromagnetic system, marked by the breaking of symmetry and the emergence of a macroscopic transverse magnetization. Magnon condensation is studied and observed in several magnetic systems, ranging from YIG films to superfluid helium 3 liquids. Our study was the first to examine magnon condensation in an atomic gas. Through position-space and momentum-space measurements of the quantum gas, we observed the magnon condensation phase transition. The transition point was characteristic of magnons that propagate with near-zero energy gap within the ferromagnetic condensate. The symmetry breaking phase transition was observed directly through

Task Progress:	spatially resolved measurements of the condensate spin. One remarkable finding was that the magnon condensation results only in a quasi-condensation, in which long-length-scale variations of the magnon condensate order parameter remain for very long evolution times. (2) Magnon interferometry in dense Bose condensed gases: As reported in our paper [Marti et al, Physical Review Letters 113, 155302 (2014)] (Ed. note: See 2015 Task Book report bibliography), we realized an interferometer in which magnon waves at several different momenta propagated coherently. These magnon waves were produced by a coherent optical excitation of a ferromagnetic spinor Bose-Einstein condensate. Specifically, we produced a pattern of magnetization containing magnons at three different momenta. The interference between these magnon waves was evident in the time variation of the density of the magnon gas, which we observed directly through spin-sensitive optical imaging. A truly remarkable feature of this magnon interferometer is that the coherent evolution is obtained even as the magnons propagated within the highly dense atomic medium of the ferromagnetic spinor gas. This stands in contrast with most other approaches to atom interferometry, where atomic density must be reduced as far as possible in order to avoid deleterious effects from interactions. However, magnon waves are immune to such interaction effects on account of the rotational symmetry of atomic interactions. (3) Magnon-evaporative cooling: In our publication [Olf et al., Nature Physics 11, 720 (2015)] (Ed. note: see 2016 Task Book report bibliography) we demonstrate the use of magnon gases for thermometry and cooling of highly degenerate Bose-Einstein gases. Precise measurement of the temperature of such degenerate gases is very difficult, requiring that one detect a small population of excited energy atoms atop a very large population of non-excited, ground-state atoms. Prior to our work, thermometry on degenerate Bose gases has been limited to temperatures that were 30% or more of the Bose-Einstein condensation transition temperature. We adopted a different approach: We added a small population of magnon excitations to the degenerate Bose gas, allowed those magnons to thermalize, and then selectively detected the magnon excitations to determine the temperature of the entire gas. By this method, we were able to measure temperatures as low as just 2% of the Bose-Einstein condensation temperature. These temperature readings confirmed that the Bose-Einstein condensed gases prepared in our experiment were at exceedingly low entropies, which is good news for efforts to use such gases to study many-body quantum physics. A byproduct of our work was the discovery of a cooling effect that occurs upon magnon thermalization. Such cooling can be implemented in experimental setups where direct evaporative cooling of the gas is difficult, e.g., for atoms in magnetic traps such as those employed in the CAL experiment on the International Space Station (ISS). (4) Quantum gas mixtures: Toward the end of the granting period, we pivoted our efforts to study quantum gas mixtures. We were interested specifically in mixtures of lithium and rubidium atoms, which had not been studied as a gas mixture in many previous experiments, and which could perhaps be used as a precursor for LiRb diatomic molecules. The study of quantum gas mixtures is among the exciting prospects for future experiments on ultracold atomic gases in free fall, for the reason that one can study these gas mixtures without having to suspend the atoms, usually differentially, against the force of gravity. After developing the ability to laser cool Li and Rb gases simultaneously and to load these gases together into both magnetic and optical traps, we conducted two significant experiments. In the first [Fang et al., Physical Review A 101, 012703 (2020)] (see Bibliography section), we measured the collision cross section between Li and Rb atoms with both atoms trapped in the F=1, mF = 1 hyperfine state. The measurement was performed by redistribution of momentum between different directions of motion, caused by collisions of Li atoms against a large bath of Rb atoms. The cross section was found to be small, which was bad news in terms of the prospects of cooling Li and Rb gases together via sympathetic cooling. Yet, the cross section was found to be significantly larger than was predicted by the best theoretical models of Li-Rb collisions. We worked with the theory group of Chris Greene, of Purdue University, to understand this discrepancy with theory. In the end, our analysis confirms that there is something amiss with the body of experimental measurements performed on Li-Rb gas mixtures. Hopefully this discrepancy will be cleared up by further measurements in various laboratories. A second experiment focused on spin-dependent interactions between Li and Rb spinor gases. Together, these gases form something known as a heteronuclear spinor Bose-Einstein gas. Such heteronuclear gases are predicted to support a wide range of interesting phenomena. In our study, we focused on spin relaxation wherein an initial spin distribution of Li atoms evolves owing to interactions with a large Rb spinor gas. From this relaxation, we were able to observe the effects of three distinctly different types of spin-dependent collisions, and then to characterize the strength of each of these processes. Our experiment represents the first complete characterization of spin-dependent collisions in a heteronuclear spinor gas, and sets an example for similar experiments to be performed on other quantum gas mixtures.
Bibliography Type:	Description: (Last Updated: 06/16/2021)
Articles in Peer-reviewed Journals	Fang F, Isaacs JA, Smull A, Horn K, Robledo-De Basabe LD, Wang Y, Greene CH, Stamper-Kurn DM. "Collisional spin transfer in an atomic heteronuclear spinor Bose gas." Physical Review Research. 2020 Jul-Sep;2(3):032054(R). https://doi.org/10.1103/PhysRevResearch.2.032054 , Aug-2020
Articles in Peer-reviewed Journals	Fang F, Wu S, Smull A, Isaacs JA, Wang Y, Greene CH, Stamper-Kurn DM. "Cross-dimensional relaxation of 7Li-87Rb atomic gas mixtures in a spherical-quadrupole magnetic trap." Physical Review A. 2020 Jan;101(1):012703. https://doi.org/10.1103/PhysRevA.101.012703 , Jan-2020